TAHIR ALI

PCB Design Engineer | Hardware Design | Firmware Development

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SUMMARY

Electrical Engineering student and freelance PCB Design Engineer with hands-on experience taking boards from schematic capture through layout, DRC/ERC verification, and manufacturing-ready output in KiCad and Altium Designer. Delivered PCB design projects for clients on Fiverr, Upwork, and Freelancer.com, alongside embedded firmware work on ESP32, STM32, and Arduino platforms. Comfortable owning a design from requirements to Gerbers.

FREELANCE EXPERIENCE

PCB Design Engineer — Freelance (Fiverr / Upwork / Freelancer.com) *2–3 years*

- Designed PCBs from scratch for clients worldwide: requirement gathering, schematic capture, component selection, PCB layout, and Gerber/BOM generation.
- Delivered manufacturer-ready files compatible with JLCPCB, PCBWay, and OSHPark.
- Worked across microcontroller dev boards, USB interface boards, and power electronics boards in KiCad and Altium Designer.

PCB DESIGN PROJECTS

Custom ESP32 Development Board *KiCad · 2-Layer PCB · ESP32-WROOM-32*

- Designed a complete 2-layer development board with USB Type-C input, ESD protection (USBLC6), and a 3.3V LDO regulator (AP2112K).
- Implemented boot/reset circuitry with RC filtering, solid ground plane for EMI reduction, and an RF antenna keep-out zone.
- Broke out UART, SPI, and I2C interfaces with full GPIO headers; verified clean with DRC and ERC.

USB to UART Converter Board *KiCad · CP2102 · USB Type-C*

- Designed a compact USB-to-UART bridge board around the CP2102 chip with a USB Type-C receptacle connector.
- Added ESD protection diodes on data lines and TX/RX status LEDs; delivered a clean, DRC/ERC-verified layout.

STM32 GPS Tracker Board *KiCad · STM32 · GPS Module*

- Designed a multi-layer board integrating an STM32 MCU, GPS module (with antenna keep-out), and SD card slot for data logging.
- Routed with a dedicated ground plane and power management circuitry; verified with DRC/ERC.

DC-DC Buck Converter PCB *KiCad · Power Electronics*

- Designed a manufacturing-ready buck converter PCB with zero ERC and zero DRC errors/warnings, including schematic capture, copper pour, and 3D visualization. View on GitHub

Water Level Indicator PCB *KiCad & Altium · 2-Layer PCB · Mar 2026*

- Designed a 2-layer PCB using BC547 transistors, LEDs, and a buzzer, from schematic through footprint assignment, layout, and 3D visualization.
- Resolved routing conflicts, footprint mismatches, and DRC errors across both KiCad and Altium Designer workflows.

EMBEDDED & FIRMWARE PROJECTS

Smart Blind Assistance System — Winner *ESP32-C6 · Dec 2025*

- Built an assistive device for obstacle detection with real-time environmental sensing and an alert system for visually impaired users, in a portable, low-power design.

STM32 UART Communication System *STM32 · Jan 2026*

- Implemented interrupt-based UART communication between an STM32 MCU and PC, controlling peripherals via serial commands.

Real-Time Edge AI Object Detection using YOLO *Raspberry Pi · Mar 2025*

- Deployed YOLO on Raspberry Pi with a live camera module for real-time object detection and bounding-box labeling; optimized for edge-device performance.

RESEARCH

Quantization-Aware Compilation on RISC-V for Edge AI Devices *LUMS (PPS) · Mar 2026*

- Presented a technical poster on optimizing AI models for edge deployment, focused on quantization techniques and RISC-V-based low-power architectures.

CERTIFICATIONS

- Mastering KiCAD: Open-Source PCB Design for Beginners — Coursera (Jul 2026). Verify

SKILLS

- PCB Design & Layout: Schematic Capture, Footprint Design, Multi-layer Routing, Impedance-aware Layout, Copper Pour, DRC/ERC Verification, Gerber & BOM Generation
- Firmware & Embedded: C, C++, Python, Microcontroller Programming, Interrupts, RTOS, UART, SPI, I2C
- Other: Circuit Analysis, Problem Solving, Team Work, Communication, Leadership

TOOLS & MICROCONTROLLERS

- Design Tools: Altium Designer, KiCad, Arduino IDE, Keil uVision, MATLAB, Espressif IDF
- Microcontrollers/Platforms: ESP32, ESP32-C6, STM32, ATmega328P, Arduino, Raspberry Pi 4

EDUCATION

Bachelor of Science in Electrical Engineering *2023 – 2027*

Namal University, Mianwali

Intermediate (Pre-Engineering) *2021 – 2022*

Cambridge International Science College, Talagang